Harish G

AI & Full-Stack Engineer | Java / Spring Boot · React · Python · LLM Agents, RAG & ML in Production

Bengaluru, India | +91 7892855850 | harishgreddy.work@gmail.com | LinkedIn | GitHub | harishg.com

Profile

Full-stack engineer who builds the AI as well as the application around it. Java 21 and Spring Boot on the backend, React on the front, PostgreSQL and Redis underneath, shipped as containers through CI. On the AI side I take models to production and report what they actually do: gradient-boosted forecasting measured out-of-sample against a mandatory baseline, retrieval that runs before inference with authorisation applied inside the retriever, and agents that call tools through declared schemas. Software Developer at CortexCraft.ai, where I authored 380 of 418 commits on our platform.

Core Skills

Languages Java 21 · Python · JavaScript / TypeScript · Kotlin · SQL · HTML / CSS
Backend Spring Boot 3.5 (Security, Data JPA, Validation, JJWT, Flyway) · FastAPI · Node / Express · Celery
Frontend React · Vite · Tailwind CSS · Jetpack Compose
Databases PostgreSQL · pgvector · MongoDB · Redis · Neo4j · SQLite / Room
AI & LLM LLM function and tool calling · config-driven tool registries · MCP-style tool hosting · prompt design · output guardrails
RAG pgvector (HNSW, cosine) · `text-embedding-004` · Hugging Face MiniLM · chunking with overlap · full-text fallback · role-scoped retrieval
ML LightGBM · PyCaret · scikit-learn · TensorFlow / Keras · PyTorch · QLoRA fine-tuning · WAPE / MASE / RMSE · conformal intervals
DevOps Docker · Kubernetes (Helm) · Terraform · GitLab CI · Jenkins · GitHub Actions · Prometheus
Testing pytest · Playwright · JUnit · coverage gates · append-only audit trails

Experience

Software Developer, CortexCraft.ai Jan 2026 – Present

- Authored 380 of 418 commits on the voice-agent platform, built as separate backend services with a React front end, PostgreSQL and Redis, behind 92 pytest suites, Playwright end-to-end tests and a four-stage GitLab CI pipeline.
- Built the agent tool layer: Gemini function declarations generated at runtime from agent configuration, a bidirectional tool-call loop and a generic HTTP tool executor, so onboarding a customer system is configuration rather than code.
- Built transcript document intelligence in Gemini JSON mode: schema-driven field extraction that coerces each value to its declared type and nulls anything off-type rather than fabricating it, plus summarisation with key points, sentiment and per-topic tags.
- Sole author of the deploy tree: three container images including a dedicated worker, a Helm chart with per-environment values, and Terraform for the managed cloud services.

Freelance Web & App Developer (Self-Employed) 2023 – 2026

Delivered full-stack and data projects end to end for clients, then supported them through deployment, fixes and upgrades.

Selected Projects

Lendly, lending platform *Java 21, Spring Boot 3.5, React, PostgreSQL, Flyway, Docker*

- Spring Boot 3.5 on Java 21 with Spring Security and JJWT, Data JPA and Bean Validation, Flyway versioned migrations against PostgreSQL and H2 for tests, a React front end, GitHub Actions CI and Docker Compose for local environments.

Demand Forecasting System *Python, PyCaret, scikit-learn, Docker, Streamlit*

- Sole author. Monthly forecasts across roughly 1,500 product series from a tiered zoo of about 150 labelled configurations, with one global model fitted across all series pooled rather than one per series, and a direct twelve-month horizon so error cannot compound.
- Scored out-of-sample on WAPE and MASE, with seasonal-naive as a mandatory baseline that ships if nothing beats it. Intervals are conformal, from model residuals rather than the variance of the data, and a GREEN / AMBER / RED certification reports that median WAPE is 16.6% and 18.7% of series meet the client's 10% target, rather than narrowing the bands until the requirement appears to be met.

Dental Clinic AI Assistant *Python, Flask, PostgreSQL/pgvector, Gemini, WhatsApp Business API*

- Chunked document RAG (800-token chunks, 100 overlap) embedded with `text-embedding-004` into pgvector, retrieved by cosine distance with a full-text fallback and role-scoped row filtering applied inside the retriever, so authorisation cannot be prompted around. Gemini function calling for booking behind a role-based tool router.

Education

B.E., Information Science and Engineering, Vivekananda Institute of Technology, Bangalore.